ed Examples: Logic Chains

system accounts into policy-relevant values

Carbon Sequestration (Mangroves)

1	2	3	4	5
Mangrove extent	NCP rate	Physical supply	Social cost of carbon	Monetary value
350 ha	**3.7 Mg CO2/ha/yr**	**1,295 Mg CO2/yr**	**$51 / tCO2**	**$66,045 / yr**

350 ha x 3.7 Mg CO2/ha/yr = 1,295 Mg CO2/yr x $51/tCO2 = $66,045/yr

Fish Provisioning (Reef Fisheries)

1	2	3	4	5
Annual catch	Market price	Gross revenue	Minus costs	Resource rent
180,000 kg	**$3.50 / kg**	**$630,000**	**- $420,000**	**$210,000 / yr**

180,000 kg x $3.50/kg = $630,000 - $420,000 costs = $210,000/yr resource rent

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system accounts into policy-relevant values

Coastal Protection (Coral Reefs)

1	2	3	4	5
Reef extent along coast	Assets behind reef	Replacement cost	Total replacement	Annualized value
12 km	**Infrastructure at risk**	**$5,000 / m**	**$60,000,000**	**$1,200,000 / yr**

12 km x 1,000 m/km x $5,000/m = $60M total, annualized at 2% = $1.2M/yr

ethods connect

mple 1)

ccount (area) multiplied by literature rate, then valued ost of Carbon. Most accessible method -- needs only d a published NCP rate.

Fish (Example 2)

Uses **catch statistics** and market data. Valued via resource rent (revenue minus intermediate costs). Requires economic data on fishing costs.

Coastal Protection (Example 3)

Uses **extent account** (reef length) and replacement co infrastructure. Annualized to produce annual service flo

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ystem services for one accounting area

Combined Services -- Single Accounting Area

tem Service	Ecosystem	Method	Physical Supply	Monet
visioning	Coral reef	Resource rent	180,000 kg/yr	
sequestration	Mangrove + seagrass	Social cost of carbon	2,334 Mg CO2/yr	
ion / tourism	Coral reef + beach	Direct expenditure	25,500 visits/yr	
protection	Coral reef	Replacement cost	12 km reef frontage	
ANNUAL ECOSYSTEM SERVICE VALUE				

L VALUE

Regulating and cultural services dominate

%) and coastal protection (43%) together account for 89% of the total annual value. These non-market services are invisible without ocean accounts. Provisioning (fish) contr
ue, yet it typically receives the most policy attention. Ocean accounts reveal the hidden economic importance of reef and mangrove conservation for tourism revenue and disa
powerful argument for Marine Spatial Planning and blue finance.

Worked E